Database Management Systems

Entity-Relationship Model

S Nagasundari

Department of Computer Science Engineering

Database Management Systems

Unit 1: Introduction to Database Management

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

Dr. Nagasundari S
Department of Computer Science and Engineering

Entity

- **What is an Entity?**
  - An entity is a distinct "thing" or "object" in the real world.
  - It's different from other objects and can be a person, course, or more.

- **Properties of an Entity:**
  - An entity has specific attributes or properties.
  - Some properties must hold unique values to tell entities apart.
- **Example - Student Entity:**
  - In a university, each student is an entity.
  - A unique “Studnt_ID" property distinguishes individuals.

E-R model

The Entity-Relationship (E-R) model is a data modeling technique designed to assist in database design.

It enables the creation of an enterprise schema that captures the logical structure of a database.

Purpose and Function: The E-R model aids in representing real-world entities, their meanings, and interactions within a conceptual schema for database design.

E-R model

- E-R diagrams provide a graphical representation of the database's logical structure. They use symbols to depict entities, relationships, and attributes, making complex relationships easier to understand.
- E-R diagrams are valued for their simplicity and clarity, making them a widely used tool in database design.
- By mapping out the entities, attributes, and relationships before implementation, E-R Diagrams help in identifying potential issues, redundancies, or inconsistencies in the database design. This leads to more efficient and optimized database structures, reducing the need for costly modifications later.
- E-R Diagrams serve as a comprehensive documentation tool that records the structure of the database.

Entity

- **Unique Identification of entity:**
  - Unique property values identify each entity.
  - For instance, a student’s “student_id" value like PES001 makes them distinct.
  - Courses can also be entities in a university setting.
  - A "course_id" attribute uniquely identifies each course.
- **Concrete and Abstract Entities:**
  - Entities can be concrete (like people or books) or abstract (like courses or reservations).
  - Both types serve as distinct objects in the system.

Entity Sets

- **Entity Set (or Entity Collection):**
  - An entity set is a group of entities of the same kind sharing common attributes or properties.
  - For instance, "instructor" represents all university instructors, and "student" represents all university students.

E1 is an entity having Entity Type Student and the set of all students is called Entity Set.

- **Extension vs. Entity Set:**
  - The real collection of entities is the "extension" of an entity set.
  - Similar to the difference between a relation and a relation instance.
- **Non-Disjoint Entity Sets:**
  - Entity sets can overlap these sorts of entity sets are called non-disjoint sets.
  - Example: "person" entity set includes instructors, students, both, or neither.

Entity Sets

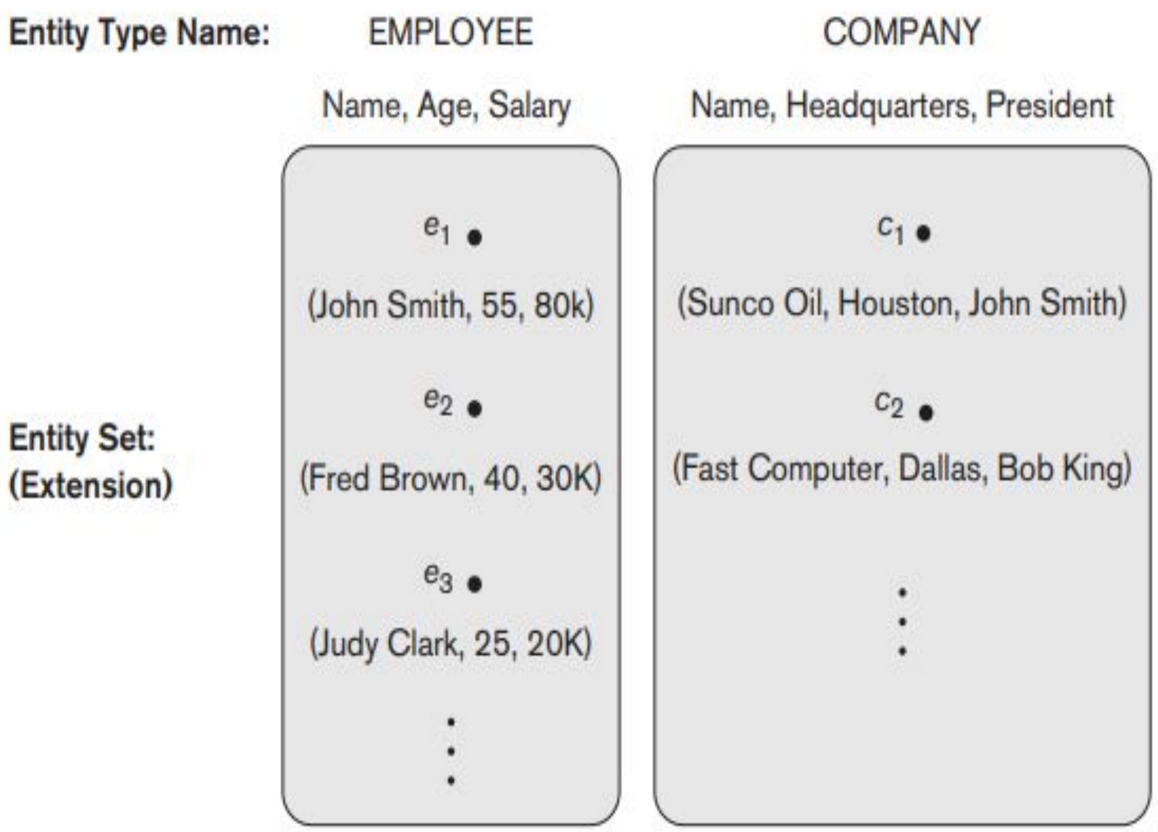
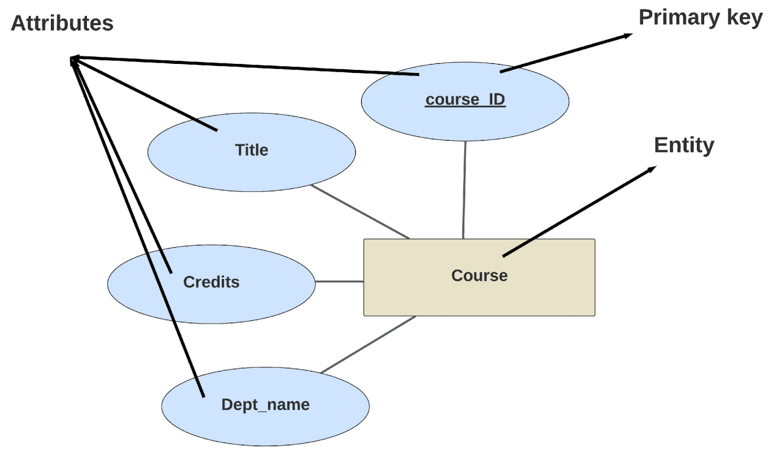


Figure 3.6
Two entity types, EMPLOYEE and COMPANY, and some member entities of each.

Entity sets

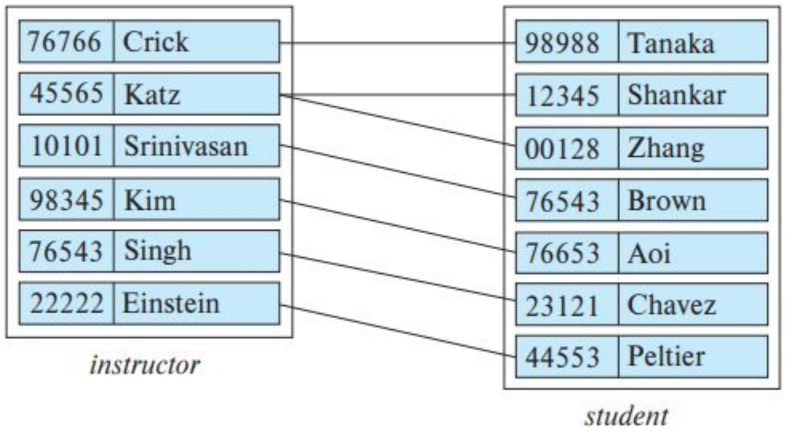
- Entity and Attribute representation:
  - An E-R diagram uses rectangles to depict entity sets, and ovals to represent the attributes.



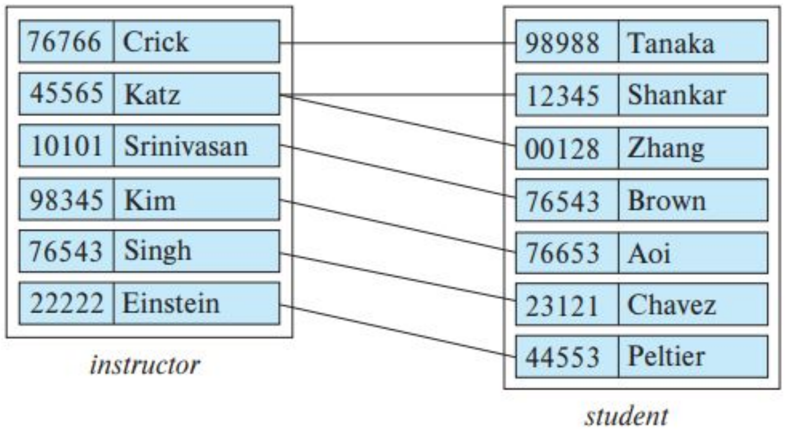
- Primary Key Indication:
  - Primary key attributes are underlined.
- For instance, let us consider a university database
  - Universities track various aspects, leading to multiple entity sets.
  - Courses is an entity set with attributes like course id, title, dept name, and credits
  - For courses the ER representation is as shown

Relationship and Relationship sets

- **What is a Relationship:**
  - A relationship signifies connections between entities.
  - Example: "advisor" links instructor Katz to student Shankar.
- **Relationship Set:**
  - A relationship set contains relationships of the same type.
  - Example: "advisor" relationship set links students and their advisors.
- **Entity Sets and Relationships:**
  - Consider "instructor" and "student" entity sets.
  - "advisor" relationship set signifies student-advisor connections.

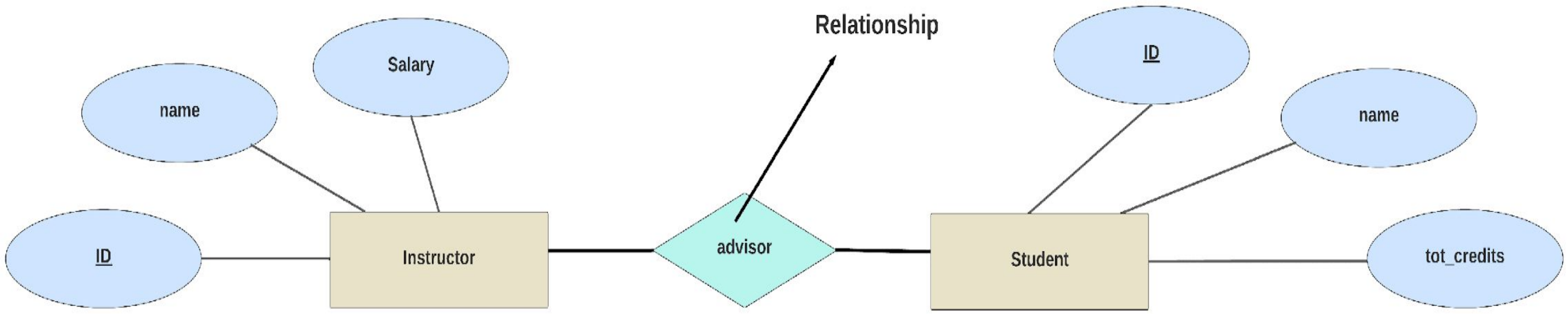


- **Relationship Instance:**
  - A relationship instance in an E-R schema represents an association between the named entities in the real-world enterprise that is being modeled.
  - As an illustration, the individual instructor entity Katz, who has instructor ID 45565, and the student entity Shankar, who has student ID 12345, participate in a relationship instance of advisor.
  - This relationship instance represents that in the university, the instructor Katz is advising student Shankar



Relationship and Relationship sets

- **Representation in E-R Diagram:**
  - In an E-R diagram, relationships are depicted by diamonds.
  - Lines connect the diamond to relevant entity sets (rectangles).
- For example let us try and represent the advisor relation



Relationship sets

- A relationship type R among n entity types E1, E2, . . . , En defines a set of associations—or a relationship set—among entities from these entity types.
- Similar to the case of entity types and entity sets, a relationship type and its corresponding relationship set are customarily referred to by the same name, R.
- Mathematically, the relationship set R is a set of relationship instances ri, where each ri associates n individual entities (e1, e2, . . . , en), and each entity ej in ri is a member of entity set Ej , $1 \leq j \leq n$.
- Hence, a relationship set is a mathematical relation on E1, E2, . . . , En; alternatively, it can be defined as a subset of the Cartesian product of the entity sets E1 × E2 × . . . × En.
- Each of the entity types E1, E2, . . . , En is said to participate in the relationship type R; similarly, each of the individual entities e1, e2, . . . , en is said to participate in the relationship instance ri = (e1, e2, . . . , en).

Recursive Relationships

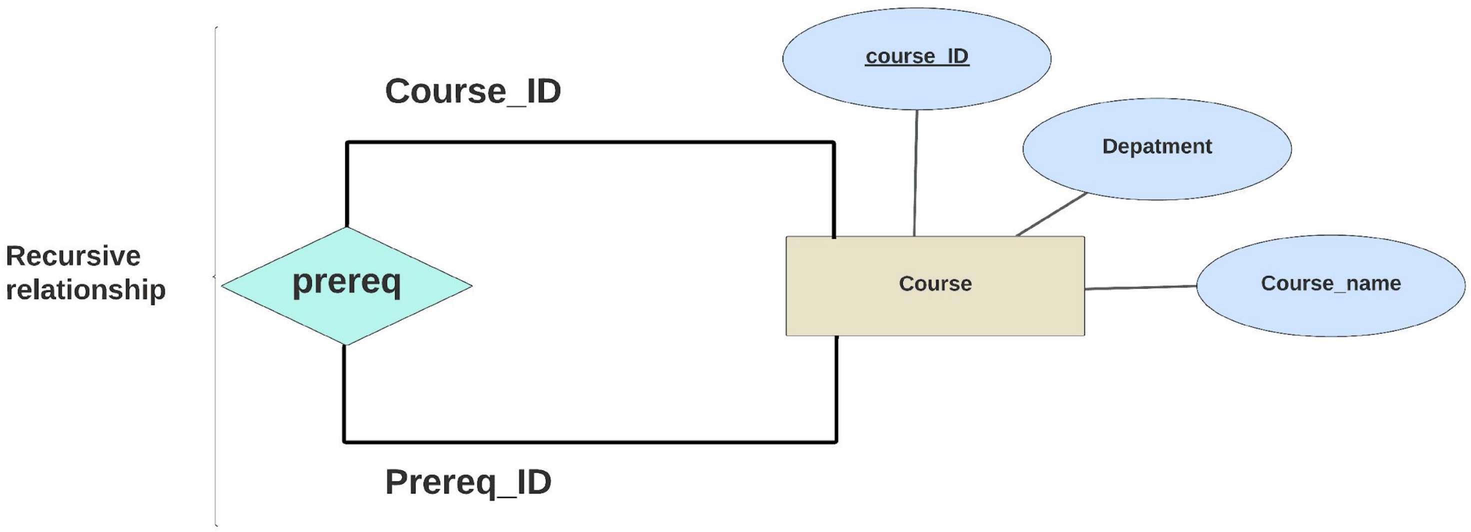
- The function that an entity plays in a relationship is called that entity's **role**.
- Since entity sets participating in a relationship set are generally distinct, roles are implicit and are not usually specified.
- However, they are useful when the meaning of a relationship needs clarification. Such is the case when the entity sets of a relationship set are not distinct; that is, the same entity set participates in a relationship set more than once, in different roles.
- In this type of relationship set, sometimes called a **recursive relationship set**, explicit role names are necessary to specify how an entity participates in a relationship instance.

Recursive Relationships

- Let us try and understand it with the help of an example
- consider the entity-set course that records information about all the courses offered in the university. Now some courses that are offered by the university have some prerequisite courses which are needed to be completed
- To depict the situation where one course (C2) is a prerequisite for another course (C1) we have a relationship set prereq that is modeled by ordered pairs of course entities.
- So how do we indicate this in the ER diagram?

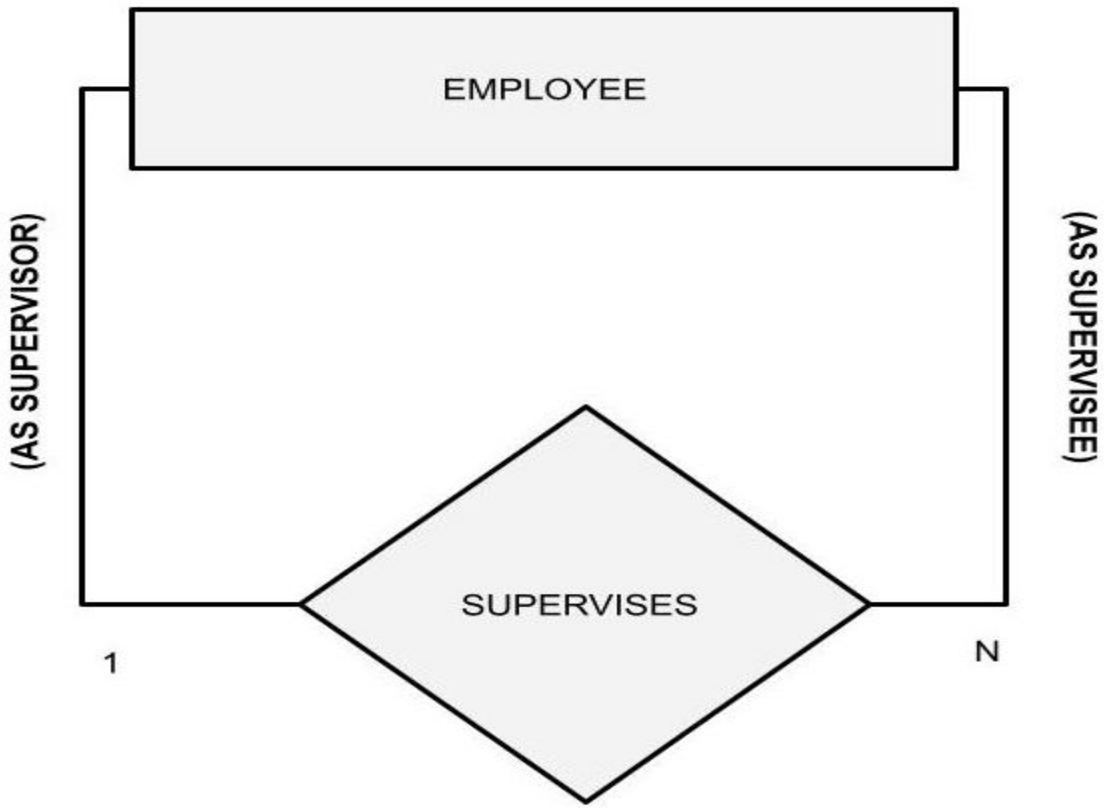
Recursive Relationships

- We indicate roles in E-R diagrams by labeling the lines that connect diamonds to rectangles.
- The diagram below shows the role indicators course id and prereq id between the course entity set and the prereq relationship set.



Recursive Relationship - Employee - Supervisor Example

Can you think of another example of a recursive-relationship?



Relationship Set and it's degree

- The relationship sets advisor is an example of a **binary relationship set**—that is, one that involves two entity sets.
- Most of the relationship sets in a database system are binary.
- Occasionally, however, relationship sets could even involve more than two entity sets. The number of entity sets that participate in a relationship set is the **degree of the relationship set**.
  - A binary relationship set is of degree 2;
  - A ternary relationship set is of degree 3.

Relationship Set Example

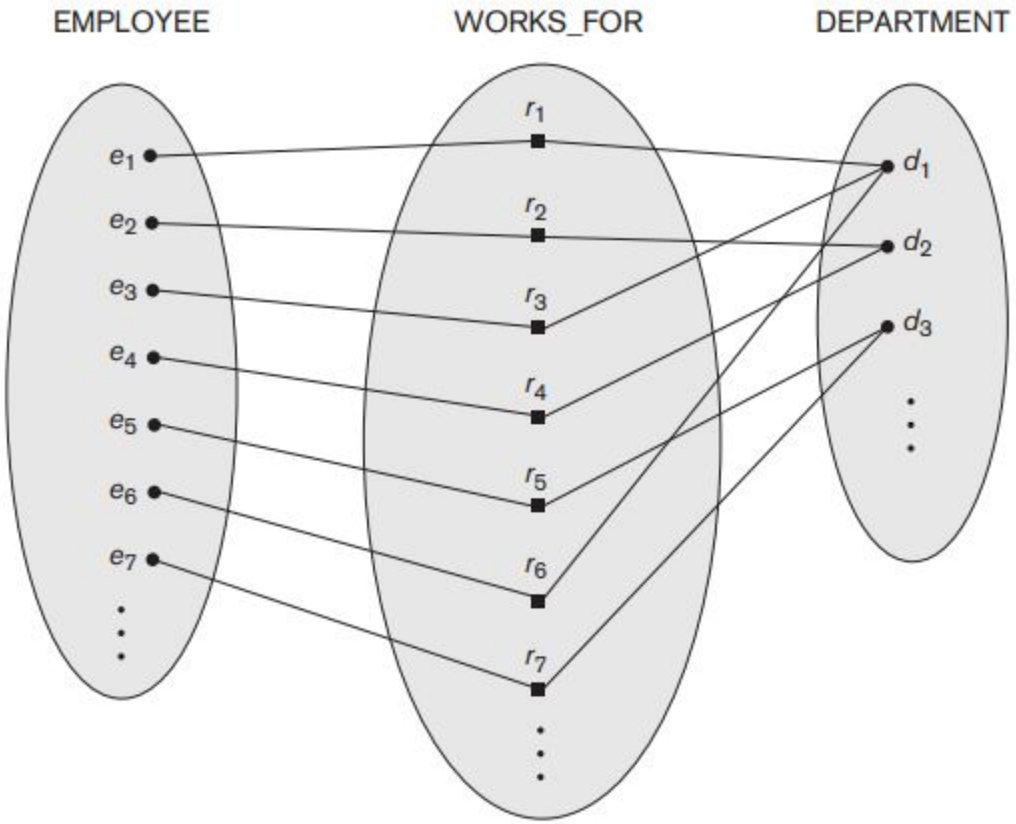


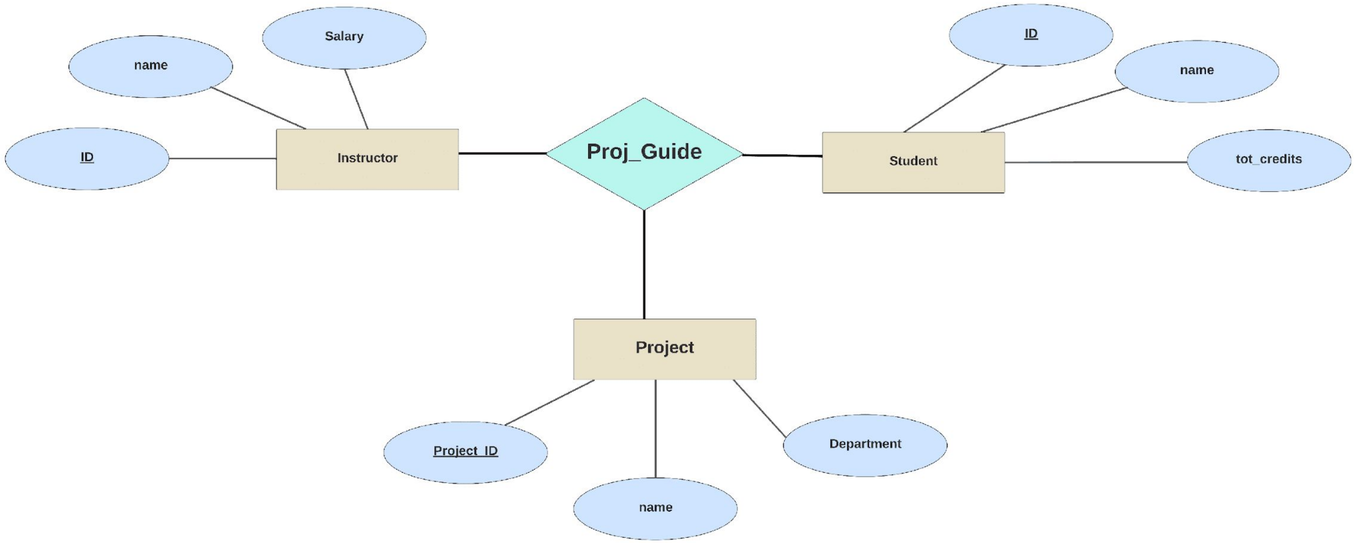
Figure 3.9
Some instances in the WORKS_FOR relationship set, which represents a relationship type WORKS_FOR between EMPLOYEE and DEPARTMENT.

Ternary Relationships

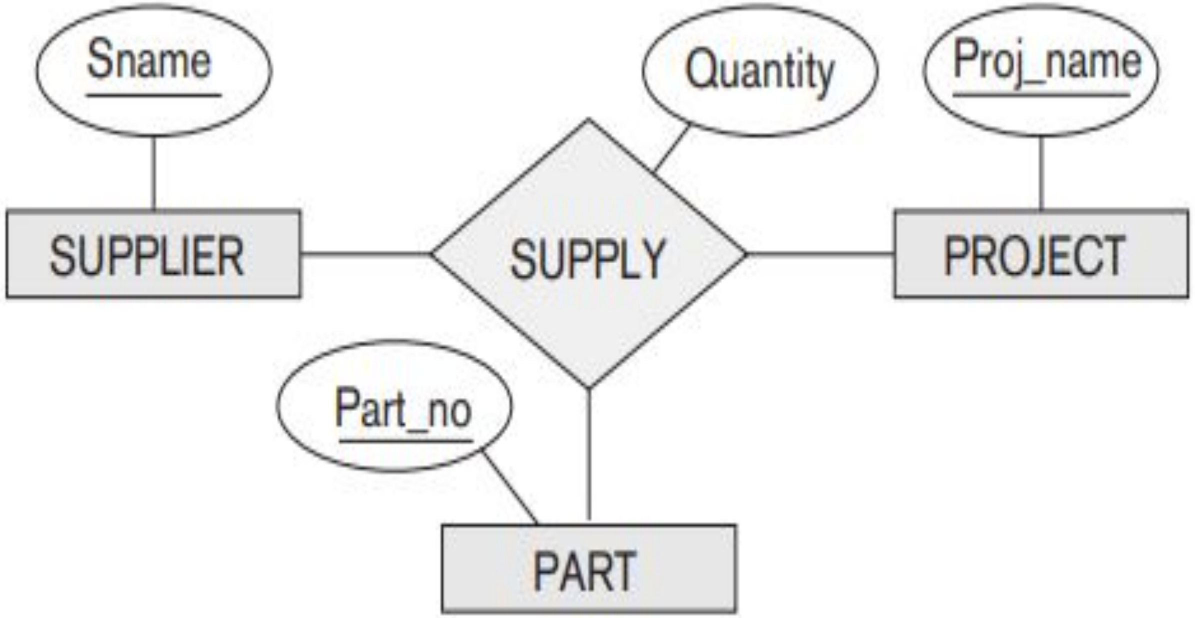
- Let us consider an example, suppose that we have an entity set project that represents all the research projects carried out in the university.
- Consider the entity sets instructor, student, and project. Each project can have multiple associated students and multiple associated instructors. Furthermore, each student working on a project must have an associated instructor who guides the student on the project.
- Suppose we were to represent the information of which instructor is guiding which student for which project
- How do we represent this or model this in an ER diagram?

Ternary Relationship Example

- To represent the above situation, we would have to relate the three entity sets through a ternary relationship set proj_guide, This ternary relation relates entity sets instructor, student, and project.
- An instance of projguide indicates that a particular student is guided by a particular instructor on a particular project.
- Note that a student could have different instructors as guides for different projects, which cannot be captured by a binary relationship between students and instructors.



Ternary Relationship Example



THANK YOU

S Nagasundari
Department of Computer Science and Engineering
nagasundaris@pes.edu